0.1 Dynamic Treatment Regimes

In many treatment and policy settings the analyst makes a sequence of decisions, the right choice at each step depends on the subject's evolving state, and the estimation target is the decision rule itself rather than a fixed treatment level. At each of K decision points the analyst observes the subject's state S_k (covariates, past treatments, past responses), chooses a treatment A_k, and observes the next state; after the final decision a scalar outcome Y is realised. The motivating example in Murphy (2003), used throughout this section, is the Fast Track prevention program, a randomized study targeted at children flagged for elevated behavioral problems in kindergarten,[1] in which the home-visiting intensity each semester was set adaptively from a six-item family-functioning score S_j, with low scores indicating greater need. The design team rejected a uniform schedule because "providing too many home visits might be detrimental, increasing risk for family dependency, pejorative labeling (by self and others) and cause attrition," and adopted an adaptive rule. The estimation target is the rule $\pi^* = (\pi_1^*, \ldots, \pi_K^*)$ that maximises the expected outcome; a standard randomized-trial analysis recovers the marginal effect of a fixed protocol, not a rule.

Notation and identification

Write $\bar{A}_k = (A_1, \ldots, A_k)$ for the treatment history through decision k, $\bar{S}_k = (S_1, \ldots, S_k)$ for the state history, and $\bar{H}_k = (\bar{S}_k, \bar{A}_{k-1})$ for the observed information just before decision A_k. For each potential treatment path $\bar{a}_K$, the potential outcome $Y^*(\bar{a}_K)$ denotes the response that would have been observed under that path (Rubin, 1978; Robins, 1986). A dynamic treatment regime $\pi = (\pi_1, \ldots, \pi_K)$ is a sequence of maps $\pi_k : \bar{H}_k \mapsto A_k$, with value $V(\pi) = \mathbb{E}[Y^*(\pi)]$ and optimization target $\pi^* \in \arg\max_\pi V(\pi)$.

Identification rests on three assumptions; Chapter **??** treats failure of the third. *Consistency* states that $Y = Y^*(\bar{A}_K)$ and $S_k = S_k^*(\bar{A}_{k-1})$, so observed outcomes equal realized potential outcomes. *Positivity* requires every action of interest to receive strictly positive probability conditional on every history that arises in the population. *Sequential ignorability* requires $A_k \perp \{Y^*, S_{k+1}^*, \ldots\} \mid \bar{H}_k$ for every k, meaning the decision A_k may depend on the observed history but not on unobserved factors that also affect future states and

[1]The program was designed to prevent the emergence of conduct disorders and drug use in at-risk children.

outcomes.

The backward recursion

Under these three assumptions, the optimal value functions satisfy the finite-horizon backward recursion

$$V_{K+1}(\bar{h}_{K+1}) = Y, \tag{1}$$

$$Q_k(\bar{h}_k, a_k) = \mathbb{E}\Big[V_{k+1}(\bar{H}_{k+1}) \,\Big|\, \bar{H}_k = \bar{h}_k,\ A_k = a_k\Big], \tag{2}$$

$$V_k(\bar{h}_k) = \max_{a_k} Q_k(\bar{h}_k, a_k), \qquad \pi^*_k(\bar{h}_k) = \arg\max_{a_k} Q_k(\bar{h}_k, a_k). \tag{3}$$

Murphy calls V_k the *optimal benefit-to-go*, inverting the engineering "optimal cost-to-go" of Bertsekas (1996) since the response is to be maximized rather than minimized. The recursion is, in her words, "a finite time version of Bellman's equation (Bellman, 1957)."

What separates the setting from textbook backward induction is identification. Engineering dynamic programming is given the joint distribution of states and outcomes indexed by every possible treatment path; for any candidate path the analyst can compute or simulate the resulting distribution and take expectations. Murphy's data are different. The cohort has N subjects each observed once along the single realized path that the treating physician chose. For any other path that the subject did not follow, the relevant $(\bar{S}_K, Y)$ distribution is counterfactual and unobserved. Worse, the empirical conditional distribution of S_{k+1} given S_k and $A_k = a$ in the cohort is not generally the distribution of S_{k+1} that would arise if every subject were assigned a at stage k, because the decision A_k may be correlated with unobserved factors that also affect S_{k+1}. Running the empirical Bellman backup on the cohort directly would absorb this bias into every $\hat{Q}_k$.

Sequential ignorability rules out the bias. Conditional on $\bar{H}_k$, the action A_k is independent of future potential outcomes, so any factor that influences both A_k and future states or outcomes must be measured in the history. The treating physician may consult anything in the chart, including past treatment responses, but does not condition on private information unavailable to the analyst. Under this assumption Robins's G-computation formula identifies the counterfactual outcome distribution from the observed-data distribution, and the recursion in (2)–(3) runs empirically. Take the

$K = 2$ case with binary actions to see what this looks like in practice. At stage two, $Q_2(\bar{h}_2, a_2) = \mathbb{E}[Y \mid \bar{H}_2 = \bar{h}_2, A_2 = a_2]$ is estimated by regressing Y on $(\bar{h}_2, a_2)$ across the cohort, giving $\hat{Q}_2$ and $\hat{V}_2(\bar{h}_2) = \max_{a_2} \hat{Q}_2(\bar{h}_2, a_2)$ at every history. At stage one, $\hat{V}_2(\bar{H}_2)$ is then treated as a synthetic outcome and regressed on $(\bar{h}_1, a_1)$, yielding $\hat{Q}_1$. The optimal rule at each stage is the argmax of the corresponding regression. The whole estimator is two regressions executed in sequence. In the offline-RL literature this backward-induction-with-regression procedure is known as Fitted Q-Iteration (Ernst et al., 2005), and it is precisely the batch form of the Bellman backup that Q-learning implements online. Sections 3 and 4 of Murphy (2003) refine the procedure by reparameterising in terms of regret contrasts, which lets the estimator place smoothness assumptions only on the quantities directly relevant for the optimal rule.

Connection to Q-learning and the dictionary

The recursion in (2)–(3) is the same Bellman backup that drives Q-learning, covered in Section ??. Watkins and Dayan (1992) estimates Q^* from a stream of (s, a, r, s') transitions, with each observation nudging the current estimate by stochastic approximation. Murphy estimates the same conditional expectation in batch, with one regression per decision stage on the full cohort. The role that successive transitions play in Watkins's setting is played by successive subjects in Murphy's. Enrolling another patient adds a row to each per-stage regression and sharpens $\hat{Q}_k$, the way an arriving transition adjusts $\hat{Q}$ in Q-learning. A Murphy cohort replayed one subject at a time, with each $\hat{Q}_k$ updated by a temporal-difference step rather than refit, would converge to the same Q^*.

Three questions ground the equivalence. First, do Murphy's batch backward regression and online Q-learning recover the same optimal regime on a tractable two-stage DTR as the cohort size grows? Second, Q-learning solves the recursion by stochastic approximation rather than by one-shot regression; how does its recovered regime improve as the cohort is replayed more times at a fixed cohort size? Third, when the state is high-dimensional and tabular Q is infeasible, do the neural-network analogues of the two estimators – Neural Fitted Q-Iteration and DQN – still recover the same optimal regime?

The tractable setting is a synthetic two-stage DTR with a five-level ordinal status $S_k \in \{1, \ldots, 5\}$, a binary action, a logistic behavior policy, and an outcome that rewards treatment when status is low and penalizes it when status is high. The high-dimensional setting holds the structural form fixed but replaces the five-level state with a $p = 10$-dimensional continuous one, an autoregressive transition with treatment effect on the first coordinate, and the same threshold interaction in the outcome.[2] Both DGPs respect sequential ignorability by construction.

Figure 1 reports the value of the recovered regime, $V(\hat{\pi})/V^*$, across the three questions. Panel (Q1) shows the tabular sample-size sweep. Murphy and online Q-learning curves overlap throughout, both approaching V^* as N grows; the small residual gap at large N is the constant-α bias of Q-learning. Panel (Q2) shows the tabular training-budget sweep at $N = 300$. Q-learning at one replay epoch sits below the converged Murphy reference; by three epochs it has caught up. Panel (Q3) shows the high-dimensional sample-size sweep. Neural Fitted Q-Iteration and DQN both rise from the behavior-policy baseline toward V^* as the cohort grows, with Neural-FQI slightly more sample-efficient at moderate N because it trains the stage-two regression to convergence before bootstrapping the stage-one regression. The equivalence survives the transition to function approximation.

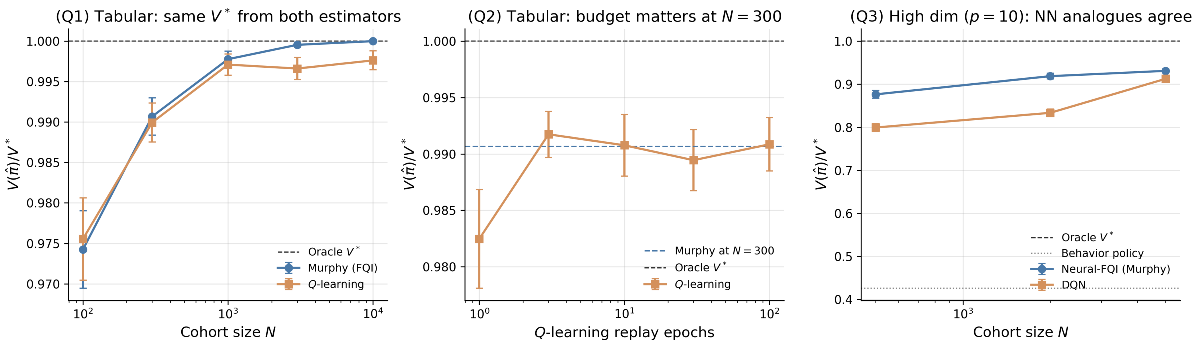

Figure 1: Murphy's batch backward regression vs online Q-learning on a synthetic two-stage DTR under sequential ignorability. (Q1) Tabular sample-size sweep at fixed Q-learning budget. (Q2) Tabular training-budget sweep at fixed $N = 300$, with Murphy's converged value as the dashed reference. (Q3) High-dimensional sweep ($p = 10$ continuous state), comparing Neural Fitted Q-Iteration to DQN with the same MLP architecture; the dotted line is the behavior-policy value. Error bars are 95% confidence intervals over Monte Carlo seeds.

Once the Bellman recursion is shared, the rest of the vocabulary aligns predictably. Table 1 translates between the two traditions following Schulte et al. (2014).[3]

[2] Implementation source: `ch10b_rl_for_ci/sims/dtr_qlearning_vs_murphy.py`. Tabular case: Q over the (history, action) grid for both estimators, oracle V^* by exact backward induction, 50 Monte Carlo seeds. High-dimensional case: two-layer MLPs with 64 hidden units for both estimators (Neural Fitted Q-Iteration fits Q stage-by-stage by full-batch MSE; DQN trains both Q networks jointly by minibatch TD), oracle V^* by Monte Carlo on the known DGP, 20 Monte Carlo seeds.

[3] Throughout the table, the RL state s is read as the full history $\bar{h}_k = (\bar{s}_k, \bar{a}_{k-1})$ at decision k, the horizon is

Table 1: Translation between dynamic treatment regimes and the g-methods of Robins (1986); Robins et al. (2000); Murphy (2003); Schulte et al. (2014) on the left, and reinforcement learning after Watkins and Dayan (1992); Sutton and Barto (2018) on the right.

Dynamic treatment regimes / g-methods	Reinforcement learning
$Q_k(\bar{h}_k, a_k)$, conditional mean of V_{k+1}	$Q(s, a)$, state-action value function
$V_k(\bar{h}_k) = \max_{a_k} Q_k(\bar{h}_k, a_k)$	$V(s) = \max_a Q(s, a)$, state value function
$d_k^{\text{opt}}(\bar{h}_k) = \arg\max_{a_k} Q_k(\bar{h}_k, a_k)$	Greedy policy $\pi^*(s) = \arg\max_a Q(s, a)$
Propensity $e_k(\bar{h}_k, a_k) = \mathbb{P}(A_k = a_k \mid \bar{H}_k = \bar{h}_k)$	Behavior / logging policy $\mu_k(a_k \mid \bar{h}_k)$
G-computation (sequential conditional regression)	Fitted-Q evaluation (Ernst et al., 2005)
IPTW for marginal structural models (Robins et al., 2000)	Off-policy importance sampling (Precup et al., 2000)
G-estimation for structural nested mean models (Robins, 2004)	Orthogonal-score off-policy evaluation (Lewis and Syrgkanis, 2021)
A-learning / contrast (blip) estimation (Schulte et al., 2014; Robins, 2004)	Advantage function (Baird, 1993)
Sequential ignorability: $A_k \perp \{Y^*, S^*_{k+1}, \dots\} \mid \bar{H}_k$	Known behavior policy $\mu_k(\cdot \mid \bar{h}_k)$ (offline RL setting)
Positivity: $e_k(\bar{h}_k, a) > 0$ for every action of interest	Coverage: $\mu_k(a \mid \bar{h}_k) > 0$ for all $a \in \operatorname{supp} \pi^*$

Two caveats apply to the translation. First, the A-learning contrast $C_k(\bar{h}_k) = Q_k(\bar{h}_k, 1) - Q_k(\bar{h}_k, 0)$ is anchored at the reference action $a_k = 0$, while the RL advantage $A_k^\pi(\bar{h}_k, a_k) = Q_k(\bar{h}_k, a_k) - V_k^\pi(\bar{h}_k)$ is anchored at the policy mean, and the two agree only when the baseline policy is constant at $a_k = 0$. Second, sequential ignorability and the offline-RL convention of a known behavior policy play the same identification role but are not literally the same statement, the former a property of the unobserved confounder structure and the latter a property of the data-collection protocol. The current chapter takes sequential ignorability as given throughout, and Chapter ?? catalogues the proxy, instrumental, sensitivity, and mediator strategies that apply when it fails.

The Murphy-Watkins bridge is now being run at scale on real electronic health records. Conservative Q-learning has been applied to vasopressor and intravenous-fluid recommendations on MIMIC-III sepsis cohorts (Kaushik et al., 2022) and to mechanical-ventilation parameter selection on MIMIC-III intensive-care patients (Kondrup et al., 2023). Cross-site policy transfer between clinical environments has been formalised as a counterfactual problem and handled with structural causal models on a sepsis simulator

finite and equal to K, all immediate rewards are zero except $R_{K+1} = Y$, and $\gamma = 1$. The Markov property of standard RL is replaced by history augmentation.

(Killian et al., 2022). Methodological-rigor work emphasises that naïve deployment is insufficient. Cross-off-policy evaluation across reward weightings identifies policies that generalise rather than overfit on intubated COVID-19 patients in the Dutch ICU Data Warehouse (Roggeveen et al., 2024), and benchmark studies show that standard offline-RL algorithms degrade markedly under realistic noise, missingness, and pharmacokinetic variability (Luo et al., 2024). Inverse-constrained methods extend the framework to infer an implicit safety specification from clinician demonstrations while the survival reward is taken as given (Fang et al., 2024). The deconfounding actor-critic of Yin et al. (2022) bakes a balancing module directly into the actor-critic learning loop, anticipating themes that recur in Chapter **??**.

0.2 Dynamic Double Machine Learning for Structural Nested Mean Models

Section 0.1 established that the recursion at the heart of g-methods and the recursion at the heart of Q-learning are the same object on history-augmented data. The natural follow-up question is what happens when the nuisance functions inside that recursion (the outcome regression, the propensity score) are estimated by modern machine learning. Lewis and Syrgkanis (2021) answer this question for the structural nested mean model. A Neyman-orthogonal version of Robins's g-estimation, combined with cross-fitting, delivers $\sqrt{n}$-consistent asymptotically normal estimates of the dynamic structural parameters, and those parameters in turn deliver off-policy evaluation at parametric rates for any target dynamic policy.

For the Fast Track cohort the structural parameter ψ_t has a direct interpretation as the marginal causal effect of an additional home visit at semester t, holding future visits at a fixed reference level. The question of this section is what valid $\sqrt{n}$ inference on ψ_1 and ψ_2 requires when the outcome and propensity nuisance functions are estimated by an off-the-shelf machine learner on the longitudinal cohort, and how to convert those structural parameters into the value of any counterfactual home-visiting regime.

Structural nested mean models

Following Robins (1986, 2004); Schulte et al. (2014), the conditional mean of the outcome given history admits a period-by-period decomposition. For each $t = 1, \ldots, T$, the *blip function* is the marginal causal effect of the period-t treatment under a fixed reference future,

$$\gamma_t(\bar{h}_t, a_t) = \mathbb{E}\Big[Y \,\Big|\, \bar{H}_t = \bar{h}_t,\ A_t = a_t,\ A_{t+1} = \cdots = A_T = 0\Big] \\ - \mathbb{E}\Big[Y \,\Big|\, \bar{H}_t = \bar{h}_t,\ A_t = 0,\ A_{t+1} = \cdots = A_T = 0\Big]. \quad (4)$$

A structural nested mean model parameterizes the blip as $\gamma_t(\bar{h}_t, a_t) = \psi_t' B_t(\bar{h}_t, a_t)$ for known basis functions B_t and unknown low-dimensional parameter $\psi_t \in \mathbb{R}^d$. The full structural parameter vector $\psi^* = (\psi_1^*, \ldots, \psi_T^*)$ identifies the value of any target dynamic policy via the g-formula applied to the partialled-out outcome (Robins, 2004).

Sequential residualisation

Lewis and Syrgkanis (2021) construct cross-fitted residuals via two nuisance regressions per period, one for the outcome and one for the treatment given history. Let $\hat{q}_t(\bar{h}_t) = \widehat{\mathbb{E}}[Y \mid \bar{H}_t]$ and $\hat{p}_{j,t}(\bar{h}_t) = \widehat{\mathbb{E}}[A_j \mid \bar{H}_t]$ for $j = t, t+1, \ldots, T$, fitted by any sufficiently fast machine learner on a held-out fold. Define the residualised outcome $\tilde{Y}_t = Y - \hat{q}_t(\bar{H}_t)$ and the residualised treatments $\tilde{A}_{j,t} = A_j - \hat{p}_{j,t}(\bar{H}_t)$. The main result of Lewis and Syrgkanis (2021), their Theorem 2, is that the structural parameter ψ_t^* satisfies the conditional Neyman-orthogonal moment restriction

$$\mathbb{E}\left[\left(\tilde{Y}_t - \sum_{j=t}^{T} \psi_j^{*\prime}\, \tilde{A}_{j,t}\right) \tilde{A}_{t,t} \,\middle|\, \bar{H}_t\right] = 0, \qquad t = T, T-1, \ldots, 1, \quad (5)$$

and the resulting system is solved recursively from $t = T$ downward. The algorithm estimates $\hat{\psi}_T$ from the period-T moment, peels off the estimated period-T effect from the outcome to define the calibrated outcome $\tilde{Y}_{T-1} - \hat{\psi}_T' \,\tilde{A}_{T,T-1}$, regresses against $\tilde{A}_{T-1,T-1}$ to obtain $\hat{\psi}_{T-1}$, and continues backward to ψ_1. Each stage solves an ordinary least squares regression on residualised data.

Asymptotic normality at $\sqrt{n}$ rates

The Neyman-orthogonal moment in Equation (5) has the crucial property that small errors in the nuisance estimates $\hat{q}_t$ and $\hat{p}_{j,t}$ enter the estimating equation at second order rather than first. Theorem 4 of Lewis and Syrgkanis (2021) shows that if each cross-fitted nuisance achieves mean-squared-error rate $o_p(n^{-1/4})$ in the population norm, then the recursive estimator $\hat{\psi}$ is $\sqrt{n}$-consistent and asymptotically normal,

$$\sqrt{n}\,(\hat{\psi} - \psi^*) \xrightarrow{d} \mathcal{N}(0, V), \tag{6}$$

with a consistently estimable covariance matrix V. The $o_p(n^{-1/4})$ condition is satisfied by Lasso regression under standard sparsity conditions, by random forests under regularity conditions on the population regression function, and by any other learner that achieves the corresponding excess-risk rate. The history $\bar{H}_t$ may itself be high-dimensional; only the structural parameter $\psi^* \in \mathbb{R}^{Td}$ is required to be low-dimensional. The resulting confidence intervals are asymptotically uniformly valid over the corresponding class of data-generating processes.

The off-policy evaluation bridge

The structural parameters ψ^* identify the value of any target dynamic policy π, because the SNMM blip representation gives a closed-form expression for the policy value $V(\pi) = \mathbb{E}[Y^*(\pi)]$ that depends only on ψ^* and the marginal action distribution under π (Lewis and Syrgkanis, 2021, Section 6). The headline claim from the paper's abstract states the bridge directly:

> *These structural parameters can be used for off-policy evaluation of any target dynamic policy at parametric rates, subject to semi-parametric restrictions on the data generating process.*

This is the bridge promised at the close of Section 0.1. Murphy's Q-recursion estimated under semiparametric efficiency gives parametric-rate inference on policy value, even when the controls inside the recursion are estimated by black-box machine learning. The simulation study in Section 0.6 demonstrates the empirical $\sqrt{n}$ coverage on a two-stage SNMM. At $n = 4000$, Dynamic DML attains 96.5% empirical coverage of nominal 95% confidence intervals on ψ_1 and 93.0% on ψ_2, with RMSE shrinking at the rate predicted by Theorem 4 of Lewis and Syrgkanis (2021). The naive panel regression that omits the time-varying confounder carries a bias of -0.19 on ψ_1 and $+0.93$ on ψ_2 that does not shrink with the sample size, and its confidence intervals cover the truth in 1% and 0% of replications respectively. MSM with stabilised inverse-probability weighting is consistent but slow; at the same sample size its coverage is 88% on ψ_1 and 73% on ψ_2.

Recursive Riesz representers

The Neyman-orthogonal moment of Lewis and Syrgkanis (2021) is hand-derived. The residual-on-residual construction in Equation (5) achieves orthogonality by inspection, because residualising both sides of a regression makes the moment locally insensitive to small perturbations in the nuisance regressions. Chernozhukov et al. (2023) automate this step. Rather than write the orthogonal moment in closed form, they characterize the de-biasing correction as a sequence of *Riesz representers* solving a system of recursive linear functional equations, and they show that these representers can be learned by minimising a sequence of quadratic Riesz losses without ever writing the orthogonal moment in closed form. Formally, define $f_{T+1}(\bar{h}_{T+1}) = Y$ and the nested mean regressions $f_t(\bar{h}_t, a_t) = \mathbb{E}\Big[f_{t+1}\Big(\bar{H}_{t+1}, \pi_{t+1}(\bar{H}_{t+1})\Big) \mid \bar{H}_t, A_t = a_t\Big]$ for $t = T, \ldots, 1$. The policy value $\theta(\pi) = \mathbb{E}[Y^*(\pi)]$ admits the debiased representation

$$\theta(\pi) = \mathbb{E}\Bigg[f_1\Big(\bar{H}_1, \pi_1(\bar{H}_1)\Big) + \sum_{t=1}^{T} \alpha_{t-1}(\bar{H}_{t-1}, A_{t-1})\Big(f_{t+1}(\bar{H}_{t+1}, \pi_{t+1}) - f_t(\bar{H}_t, A_t)\Big)\Bigg], \quad (7)$$

where $\alpha_0 \equiv 1$ and each $\alpha_t : \mathcal{H}_t \times \mathcal{A}_t \to \mathbb{R}$ is the Riesz representer of the linear functional $L_t(g) = \mathbb{E}\Big[\alpha_{t-1}(\bar{H}_{t-1}, A_{t-1})\, g(\bar{H}_t, \pi_t)\Big]$. Each α_t is estimated by minimising the quadratic Riesz loss $\mathbb{E}_n\Big[\alpha^2(\bar{H}_t, A_t) - 2\,\hat{\alpha}_{t-1}(\bar{H}_{t-1}, A_{t-1})\,\alpha(\bar{H}_t, \pi_t)\Big]$, with no closed-form inverse-propensity-weight expression required. The two constructions are complementary. Lewis and Syrgkanis (2021) target the SNMM blip parameters at parametric rates; Chernozhukov et al. (2023) target the policy value $\theta(\pi)$ non-parametrically. Both achieve the mixed-bias product structure under which $o_p(n^{-1/4})$ nuisance error suffices for the target oracle rate.[4]

In offline-RL vocabulary, the Lewis-Syrgkanis estimator is the orthogonal-score version of *fitted-Q evaluation*, with the residual-on-residual moment in Equation (5) replacing the ordinary mean-squared-error loss that FQE uses to fit the period-t value function. The recursive Riesz representer construction of Chernozhukov et al. (2023) is *automatic debiased off-policy evaluation*, applicable to any RL OPE estimator that targets a smooth functional of a sequence of conditional means, including fitted-Q evaluation, model-based

[4]The reason this works at the population-risk level is the orthogonal statistical learning theory of Foster and Syrgkanis (2023), whose Theorem 1 shows that under prediction-space strong convexity of the target loss and Neyman orthogonality of the loss derivative, the excess risk of a two-stage cross-fitted estimator has nuisance error entering at second order, so $n^{-1/4}$ mean-squared error on each nuisance suffices for the target oracle rate n^{-1}.

OPE, and marginalised importance sampling. The bridge in this section is what lets those RL OPE methods inherit the $\sqrt{n}$ rate that semiparametric efficiency theory establishes for the SNMM target.

0.3 Dynamic Offline Policy Learning

Section 0.2 delivers $\sqrt{n}$ confidence intervals on dynamic structural parameters under sequential ignorability, which in turn deliver off-policy evaluation at parametric rates for any target dynamic policy. This subsection turns from evaluation to optimization. Given observational data on n trajectories, can we *learn* an optimal dynamic regime $\pi^* \in \arg\max_{\pi \in \Pi} V(\pi)$ at the same $\sqrt{n}$ regret rate, using modern machine learning for the nuisances? Sakaguchi (2024) answers yes, by combining the doubly-robust AIPW machinery of Athey and Wager (2021) with fitted-Q-evaluation in a backward-induction loop.

For the Fast Track cohort, a natural restricted policy class is the family of threshold rules in the family-functioning score, $\pi_t(\bar{h}_t) = \mathbb{1}\{S_{t,j} < c_t\}$ for thresholds (c_1, c_2) to be learned from data. The question of this section is the welfare-maximising thresholds with a $\sqrt{n}$-rate welfare-regret guarantee, when the cohort is observational and the nuisances are estimated by machine learning.

The static $T = 1$ precursor

In the single-period treatment-choice problem the offline policy-learning literature is well-developed. Kitagawa and Tetenov (2018) introduced empirical welfare maximization with known propensities and proved a matching minimax bound of $\Theta(\sqrt{\mathrm{VC}(\Pi)/n})$ on welfare regret, applied to the National JTPA Study. Athey and Wager (2021) generalized to unknown propensities via cross-fitted doubly-robust scores (their Equation 14), maintaining the $O_p(\sqrt{\mathrm{VC}(\Pi)/n})$ rate under product-rate nuisance conditions and extending to instrumental-variable identification. Zhou et al. (2023) extended further to multi-action treatments with an exact decision-tree-search algorithm. All three are the $T = 1$ special case of what follows.

Backward-induction AIPW under sequential ignorability

For each stage $t = 1, \dots, T$, fix a policy class Π_t over which to optimize the stage-t rule, and let $\Pi = \Pi_1 \times \cdots \times \Pi_T$. Define the action-value function for any candidate future policy $\pi_{(t+1):T} = (\pi_{t+1}, \dots, \pi_T)$ recursively by

$$Q_t^{\pi_{(t+1):T}}(\bar{h}_t, a_t) = \mathbb{E}\Big[Y_t + Q_{t+1}^{\pi_{(t+2):T}}\Big(\bar{H}_{t+1}, \pi_{t+1}(\bar{H}_{t+1})\Big) \,\Big|\, \bar{H}_t = \bar{h}_t, A_t = a_t\Big], \tag{8}$$

with terminal condition $Q_T(\bar{h}_T, a_T) = \mathbb{E}[Y_T \mid \bar{H}_T = \bar{h}_T, A_T = a_T]$. The recursion estimates Q in the form of *fitted-Q-evaluation* (Ernst et al., 2005) applied under sequential ignorability. Let $e_t(\bar{h}_t, a_t) = \mathbb{P}(A_t = a_t \mid \bar{H}_t = \bar{h}_t)$ denote the stage-t propensity.

Sakaguchi (2024) estimates the optimal regime $\hat{\pi} = (\hat{\pi}_1, \dots, \hat{\pi}_T)$ by backward induction with cross-fitted nuisances. Randomly partition the data into K folds, and let $\hat{Q}_t^{-k}$ and $\hat{e}_t^{-k}$ denote the nuisance estimators fitted on the complement of the k-th fold (any sufficiently fast machine learner). Starting at $t = T$, construct the cross-fitted AIPW score for any candidate stage-T policy π_T,

$$\hat{\Gamma}_{i,T}(a) = \hat{Q}_T^{-k(i)}(\bar{H}_{i,T}, a) + \frac{\mathbb{1}\{A_{i,T} = a\}}{\hat{e}_T^{-k(i)}(\bar{H}_{i,T}, a)}\Big(Y_{i,T} - \hat{Q}_T^{-k(i)}(\bar{H}_{i,T}, A_{i,T})\Big), \tag{9}$$

and select $\hat{\pi}_T = \arg\max_{\pi_T \in \Pi_T} \frac{1}{n}\sum_{i=1}^n \hat{\Gamma}_{i,T}(\pi_T(\bar{H}_{i,T}))$. Given $\hat{\pi}_T$, refit $\hat{Q}_{T-1}^{\hat{\pi}_T}$ by fitted-Q-evaluation against the calibrated next-stage value, build the stage-$T-1$ AIPW score from Equation (9) with the new Q-function in the position of $\hat{Q}_T^{-k(i)}$, and select $\hat{\pi}_{T-1}$. Continue backward to $\hat{\pi}_1$. Each stage solves a single classification-style policy optimization on cross-fitted scores; the nuisances at stage t depend only on the previously estimated future policies $\hat{\pi}_{(t+1):T}$, never on the candidate π_t being optimized.

The $\sqrt{n}$ welfare regret bound

Sakaguchi (2024, Theorem 4.3) establishes that under sequential ignorability, overlap (Assumption 2.3 of the paper), a correct-specification condition on the policy classes (Assumption 3.1), uniform MSE convergence of $o_p(n^{-1/4})$ on both the Q-functions for the estimated future policies and the propensity scores, and an entropy-integral complexity bound on Π, the welfare regret of the estimated regime satisfies

$$V(\pi^*) - V(\hat{\pi}) = O_p\Big(\kappa(\Pi)\, n^{-1/2}\Big)\,, \tag{10}$$

where $\kappa(\Pi)$ is the entropy integral of the DTR class. This matches the $\sqrt{n}$ minimax-optimal rate that Kitagawa and Tetenov (2018) established for the static $T = 1$ case, and is the first such result for the multi-stage problem with purely nonparametric ML nuisances. A companion result (their Theorem 5.1) shows that the alternative simultaneous-optimization approach attains the same rate without requiring correct specification of the policy class, at the cost of computational tractability that degrades quickly in T.

Empirical application

Sakaguchi (2024, Section 7) applies the method to the Tennessee Project STAR class-size experiment. The treatment at each of two stages (kindergarten, first grade) is the choice between a regular-size class with a full-time teacher aide and a small-size class without one; the welfare objective is the percentile rank of combined reading and mathematics test scores at the end of first grade. Decision trees of depth one and two are used for the stage-specific policy classes, with cross-fitted random-forest nuisances. The estimated optimal dynamic regime improves academic achievement by approximately 8% relative to uniformly assigning all students to classes with a teacher aide, and by approximately 1% relative to uniformly assigning to small classes, both estimated via five-fold cross-validation. A companion field experiment in Japan (Ida et al., 2024) applies a related dynamic-targeting estimator to energy-saving rebate programs and confirms empirically that two-period dynamic targeting outperforms one-shot static targeting on realized welfare gains.[5]

In offline-RL vocabulary, the Sakaguchi estimator is the doubly-robust analog of *fitted-Q iteration* for policy optimisation. The action-value recursion in Equation (8) is fitted-Q evaluation under sequential ignorability; the augmentation in Equation (9) adds the

[5]A parallel literature targets related aspects of the same problem. Nie et al. (2021) develop an advantage doubly-robust estimator for the "when-to-treat" subclass of dynamic regimes and prove the first dynamic analog of the Athey and Wager (2021) static regret bound. Shi et al. (2024) extend to infinite-horizon offline reinforcement learning, showing that the advantage function is smoother than the Q-function and yielding a strictly faster minimax rate than fitted-Q-iteration. Luckett et al. (2020) introduce V-learning for infinite-horizon mobile-health regimes, bypassing Q-estimation by directly optimizing the value function via the Bellman estimating equation.

importance-weighted residual correction that makes the score estimator doubly robust. The connection extends downstream. Shi et al. (2024) target the advantage function directly rather than the action-value function and obtain a faster minimax rate, in the same spirit that *advantage-weighted regression* and *implicit Q-learning* replace the Q-function with smoother surrogates for offline policy improvement. The DR-AIPW scoring is the principal innovation that distinguishes the Sakaguchi guarantee from those purely-RL offline methods, which lack the doubly-robust property and so require correct specification of either the outcome model or the propensity to attain the parametric rate.

0.4 Causal Bandits

Sections 0.2 and 0.3 consider multi-stage problems where the state evolves under the influence of past treatments. This subsection collapses the horizon to a single stage but expands the structure of the action set, replacing the opaque arms of a standard multi-armed bandit with interventions on a known causal graph. The resulting "causal bandit" setting was introduced by Bareinboim et al. (2015) for the case of unobserved confounders between the agent's choice and the reward, and formalized into a regret theorem by Lattimore et al. (2016).

Bandits under unobserved confounding

The motivating example of Bareinboim et al. (2015) is a "greedy casino" with two slot machines M_1 and M_2. An unobserved binary variable D (the player is drunk) and an unobserved binary variable B (the chosen machine is blinking) jointly determine the gambler's natural arm choice $X \in \{M_1, M_2\}$ via a structural equation $X = f(D, B)$. Both confounders also affect the reward Y. The casino parameters are chosen so that the observational distribution and the interventional distribution are both flat,

$$\mathbb{P}(Y = 1 \mid X = M_j) = \mathbb{P}(Y = 1 \mid \mathrm{do}(X = M_j)) = \tfrac{1}{2}, \qquad j \in \{1, 2\}. \tag{11}$$

Bareinboim et al. (2015) run ϵ-greedy, UCB1, Thompson Sampling, and EXP3 on this environment and observe that all four are statistically indistinguishable from random guessing, with linear cumulative regret. The reason is that both the observational and interventional distributions average out the contribution of the confounders, leaving no signal at all in the average payoff per arm.

The fix proposed in Bareinboim et al. (2015) is to refine the decision objective from the marginal interventional value to the *effect of treatment on the treated*, that is, to condition on the agent's own natural intuition $X = x$ and ask what the payoff would have been under an alternative action,

$$a^* = \arg\max_a \mathbb{E}\Big[Y_{X=a} = 1 \,\Big|\, X = x\Big]. \tag{12}$$

The authors call this the *Regret Decision Criterion*. Equation (12) is generally not identified from observational and interventional data alone, but in the binary case it can be recovered through algebraic combinations of $\mathbb{P}(Y \mid X)$ and $\mathbb{P}(Y \mid \mathrm{do}(X))$, and in the general case through intention-specific randomization that interrupts the agent before action selection. Plugging this counterfactual quantity into Thompson Sampling yields Causal Thompson Sampling (TS_C), which seeds Beta posteriors from observational data via the consistency axiom and biases sampling toward the agent's intuition arm. In the greedy casino, TS_C achieves bounded cumulative regret while standard Thompson Sampling grows linearly.

Causal-graph-aware exploration

Lattimore et al. (2016) formalize the gain from exploiting causal structure as a regret theorem. They study the parallel-bandit special case: N binary parents $X_1, \ldots, X_N$ of a binary reward Y, with action set $\mathcal{A} = \{\mathrm{do}()\} \cup \{\mathrm{do}(X_i = j) : i \in [N], j \in \{0, 1\}\}$. After each action the learner observes both Y and the realized values of all non-intervened parents. Their Algorithm 1 spends half the budget on do() to estimate the marginal probabilities $q_i = \mathbb{P}(X_i = 1)$ and uses the post-action observations to credit-assign across all arms. The remaining budget is allocated to the actions whose direct-observation probability is smallest, with the cutoff τ chosen to optimize the worst-case regret. Define the graph-derived hardness

$$m(q) = \min_{\tau \in \{2,\ldots,N\}} \max\Big(\tau, \, \Big|\{i : \min(q_i, 1 - q_i) < 1/\tau\}\Big|\Big) \in [2, N]. \tag{13}$$

Theorem 1 of Lattimore et al. (2016) gives the simple-regret guarantee for Algorithm 1,

$$R_T(\text{Algorithm 1}) = \tilde{O}\left(\sqrt{m(q)/T}\right), \tag{14}$$

and Theorem 2 establishes the matching lower bound $R_T = \Omega(\sqrt{m(q)/T})$ over all algorithms. Standard best-arm-identification procedures that ignore the graph achieve $\Omega(\sqrt{N/T})$ (Lattimore et al., 2016, Theorem 4), so the graph-aware algorithm strictly improves whenever $m(q) < N$ — equivalently, whenever the parent distribution has at least one extreme coordinate. In the balanced case $q = (\frac{1}{2}, \ldots, \frac{1}{2})$ the hardness collapses to $m(q) = 2$ regardless of N; in the degenerate case $q = (0, \ldots, 0)$ the graph provides no help and $m(q) = N$. For general directed acyclic graphs Algorithm 2 of Lattimore et al. (2016) uses a truncated importance-weighted estimator with a mixing distribution η over interventions, achieving simple regret $\tilde{O}(\sqrt{m(\eta)/T})$ where $m(\eta)$ is the analogous graph-derived hardness; for the parallel bandit case $m(\eta) = m(q)$.

In reinforcement-learning vocabulary, the MABUC instance is *causal Thompson sampling*, in which the Beta posterior is indexed not by arm alone but by the agent's natural intuition $X = x$ that conditions the counterfactual reward distribution, and the Regret Decision Criterion is the Bayesian-bandit analog of an advantage estimate under unobserved confounding. The parallel-bandit Algorithm 1 is *graph-aware UCB*, a deterministic budget split that exploits the post-action observations of non-intervened parents as additional samples for arms that were not pulled. The dictionary entry for *Bandits with Unobserved Confounders* in Table 1 translates these primitives one-for-one with their RL counterparts.

The simulation study in Section 0.6 reproduces the $\sqrt{m/T}$ rate for Algorithm 1 across the hardness grid $m \in \{2, 8, 24, 48\}$ at $N = 50$, contrasted with the $\sqrt{N/T}$ floor of a graph-blind best-arm-identification baseline, and includes the greedy-casino instance of Bareinboim et al. (2015) with TS_C to demonstrate the value of causal reasoning under unobserved confounders.

0.5 Adaptive Experimentation and Post-Experiment Inference

The causal-bandit setting of Section 0.4 fixes the data-generating process and the action set and asks how to learn the best intervention. This subsection considers the dual concern. When the data themselves are collected adaptively by a Thompson-sampling-style assignment rule that the experimenter designs, what does optimal design look like, and how should the analyst conduct valid statistical inference on the resulting data? Kasy and Sautmann (2021) answer the design question; Hadad et al. (2021) answer the inference question. The two papers address different parts of the same applied workflow: design a multi-wave adaptive experiment that ends in a policy choice, then deliver confidence intervals on the welfare gain.

Design: exploration sampling for policy choice

Kasy and Sautmann (2021) consider a finite-horizon adaptive experimental design problem with K candidate treatments, T waves of approximately equal size, and a welfare objective equal to the average outcome of the policy chosen at the end of the experiment. Unlike the in-sample welfare objective targeted by classical bandit algorithms, this objective values only the post-experiment policy, not the participants' realizations during the experiment. The authors prove this distinction matters. Standard Thompson sampling, which assigns each treatment d at wave t with probability equal to the posterior probability p_t^d that d is optimal given history, is not rate-optimal for policy choice, because Thompson sampling concentrates on a single top treatment fast enough to starve the suboptimal treatments of the samples needed to verify their suboptimality.

The proposed correction is *exploration sampling*. Replace the Thompson assignment probability p_t^d with a concave transformation that flattens at the top,

$$q_t^d = S_t \cdot p_t^d \cdot (1 - p_t^d), \tag{15}$$

where S_t is a normalizing constant. Two consequences follow. First, no treatment ever receives more than 50% of the sample at any wave, regardless of how confident the posterior becomes. Second, in large samples the assignment shares converge to a non-random vector ρ^d for which the posterior probability of suboptimality decays at the same exponential rate for every suboptimal treatment; power for rejection is equalized. Theorem 1 of Kasy and Sautmann (2021) establishes that exploration sampling attains the best exponential rate of expected policy regret subject to the constraint that the top treatment receives at most half the sample, a constraint that the paper's Proposition 2 shows loses at most a factor of two relative to the fully optimal allocation.

The empirical anchor is a field experiment with Precision Agriculture for Development (PAD) in India, deploying exploration sampling across 17 waves of approximately 600 farmers each, comparing six recruitment strategies for an agricultural extension service. The realized assignment trajectory matches the theoretical prediction. The top treatment receives close to but not above half the sample, and posterior probabilities for suboptimal treatments converge to zero at comparable rates. The full experiment identifies the optimal recruitment strategy with posterior probability above 0.75 after roughly 10,000 participants.

Inference: adaptively-weighted AIPW after the experiment

Once the experiment in Kasy and Sautmann (2021) has run, the analyst still needs valid confidence intervals on the welfare of the chosen policy and on the contrasts between treatments. This is harder than it looks. Hadad et al. (2021) show that the obvious estimators all fail. The sample mean of Y_t within each arm is biased downward whenever a temporary downward fluctuation in the early phase of the experiment causes that arm to be sampled less in later phases. The inverse-propensity-weighted estimator $\hat{Q}^{\mathrm{IPW}}(w) = T^{-1}\sum_t \mathbb{1}\{W_t = w\}Y_t/e_t(w)$ is unbiased but heavy-tailed, because the inverse weights $1/e_t(w)$ grow without bound as the bandit algorithm down-weights an arm. The vanilla augmented inverse-propensity score

$$\hat{\Gamma}_t^{\mathrm{AIPW}}(w) = \hat{m}_t(w) + \frac{\mathbb{1}\{W_t = w\}}{e_t(w)}\big(Y_t - \hat{m}_t(w)\big) \tag{16}$$

inherits the same heavy-tailed behavior when averaged uniformly across time, because the conditional variances of its components depend on $1/e_t(w)$ and can diverge as the bandit assignment probabilities approach zero.

The solution proposed in Hadad et al. (2021) is to non-uniformly average the AIPW scores using history-adapted evaluation weights $h_t(w)$ that compensate for the propensity decay. The *adaptively-weighted AIPW estimator* is

$$\hat{Q}^{\mathrm{AW}}(w) = \frac{\sum_{t=1}^{T} h_t(w)\,\hat{\Gamma}_t^{\mathrm{AIPW}}(w)}{\sum_{t=1}^{T} h_t(w)}. \tag{17}$$

Their Theorem 2 establishes that for any history-adapted weights satisfying an infinite-sampling condition, a variance-convergence condition, and a Lyapunov-type moment condition, the studentized statistic is asymptotically standard normal. The simplest choice that satisfies these conditions and approximately minimizes asymptotic variance is the constant-allocation-rate weight $h_t(w) \propto \sqrt{e_t(w)}$, which compensates for propensity decay at exactly the right rate. The empirical demonstration in their Section 4 reports near-nominal 95% coverage of confidence intervals constructed from the studentized statistic under Thompson-sampling data collection, while the sample mean and the IPW estimator both fail.[6]

Together, Kasy and Sautmann (2021) and Hadad et al. (2021) close the loop for adaptive experimentation. The first paper says what assignment rule maximizes the welfare of the post-experiment policy choice; the second paper says how to construct valid confidence intervals on the resulting welfare estimates. Both treat the assignment mechanism as a Thompson-family bandit, and both deliver formal optimality results: rate-optimality for the design (Kasy and Sautmann, 2021, Theorem 1) and asymptotic normality for the inference (Hadad et al., 2021, Theorem 2). The use of an RL algorithm at the design stage of a field experiment, with causal-inference guarantees at the inference stage, is the cleanest example in this chapter of the RL-for-CI narrative bridging from theory to deployed practice.

In reinforcement-learning vocabulary, exploration sampling is a *welfare-objective Thompson-sampling variant* that trades in-sample cumulative-regret optimality for post-experiment power equalisation across suboptimal arms. The adaptively-weighted AIPW estimator is the *propensity-decay-robust off-policy-evaluation estimator* that any reinforcement-learning pipeline collecting adaptive data needs in order to recover valid inference from its own logged data, whether the adaptive collection is a contextual bandit, an online RL algorithm with an off-policy correction, or a preference-elicitation loop for RLHF.

[6] Bibaut et al. (2021) extend the adaptively-weighted AIPW framework to contextual adaptive experiments, where the assignment rule may depend on per-unit covariates. Their CADR estimator inversely weights each canonical-gradient term by an adaptively estimated conditional standard deviation, producing the first asymptotically-normal post-bandit estimator that works with contextual bandit data. The construction requires no outcome-model consistency, only that the conditional-standard-deviation estimator be measurable with respect to past data.

0.6 Simulation Studies

This subsection presents two computational experiments that ground the chapter's central claims. The first targets Section 0.2: dynamic double machine learning on a two-stage structural nested mean model delivers $\sqrt{n}$ confidence intervals on the dynamic treatment effect parameters, while a naive panel regression and a marginal-structural-model with stabilized inverse-probability-of-treatment weights both fail. The second targets Section 0.4: on the parallel-bandit family the graph-aware causal-bandit algorithm of Lattimore et al. (2016) attains $\tilde{O}(\sqrt{m(q)/T})$ simple regret while a graph-blind best-arm-identification baseline stays at the $\sqrt{N/T}$ floor, and on the greedy-casino MABUC instance of Bareinboim et al. (2015) causal Thompson sampling achieves bounded cumulative regret while standard Thompson sampling grows linearly.

Sim 1: dynamic DML on a 2-stage SNMM

The data-generating process follows a two-stage Markovian setup with high-dimensional state, confounded propensities, and treatment-confounder feedback. State dimension $p = 20$ with sparse outcome coefficients on $s = 5$ coordinates. At stage one, $X_1 \sim \mathcal{N}(0, I_p)$ and the binary treatment $T_1 \sim \text{Bernoulli}(\sigma(\gamma^\top X_1))$ with $\|\gamma\|_2 = 1.5$ supported on the first s coordinates. The period-1 state shock $\eta_1 \sim \mathcal{N}(0, \sigma_\eta^2 I_p)$ with $\sigma_\eta = 0.6$ drives the next state $X_2 = BX_1 + \alpha T_1 + \eta_1$ with $\|B\|_{\text{op}} = 0.5$ and $\alpha = e_1$, after which $T_2 \sim \text{Bernoulli}(\sigma(\gamma^\top X_2))$. The outcome

$$Y = \psi_1^* T_1 + \psi_2^* T_2 + \mu^\top X_1 + \nu^\top \eta_1 + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 0.5^2), \tag{18}$$

has true structural parameters $(\psi_1^*, \psi_2^*) = (1.0, 0.5)$ and the outcome-coupling vector $\nu = 2.0 \cdot \gamma/\|\gamma\|$ is aligned with the propensity direction, so that η_1 enters Y and also drives the period-2 propensity through X_2. This is the canonical treatment-confounder-feedback channel: conditional on (X_1, T_1), T_2 is correlated with Y's residual through η_1.

Three estimators are compared. *Naive OLS* regresses Y on (T_1, T_2, X_1), omitting the time-varying confounder X_2. *MSM with stabilized IPTW* estimates the propensity at each stage by L1-regularized logistic regression, constructs stabilized weights as the product of marginal-over-conditional propensity ratios at each stage, and fits a weighted least-squares regression of Y on (T_1, T_2). *Dynamic DML* implements Algorithm 1 of Lewis and Syrgkanis (2021): at each stage cross-fitted Lasso outcome regressions and L1-logistic propensity models on five folds, residualized at the appropriate history, with $\hat{\psi}_2$ recovered from the period-2 residual-on-residual moment and $\hat{\psi}_1$ from the peeled-off period-1 moment per Theorem 2 of the paper. Standard errors use the Z-estimator influence function $\hat{\sigma}^2(\hat{\psi}_t) = \mathrm{Var}(\tilde{T}_{t,t}\,\hat{r}_t)/\hat{J}_{t,t}^2$. Two hundred Monte Carlo replications per configuration over the sample size grid $n \in \{250, 500, 1000, 2000, 4000\}$.

Table 2 and Figure 2 report the results.[7] The naive panel regression carries a bias of approximately 0.93 on $\hat{\psi}_2$ that does not shrink with n, and its 95% confidence intervals never cover the truth. The MSM estimator is asymptotically consistent but converges slowly: its bias on $\hat{\psi}_2$ falls from 0.68 at $n = 250$ to 0.15 at $n = 4000$, but coverage of stabilized-weight sandwich confidence intervals stays well below nominal across the grid. Dynamic DML achieves the asymptotic guarantee. Bias falls from 0.06 to less than 0.005 on $\hat{\psi}_1$ and from 0.03 to 0.001 on $\hat{\psi}_2$. RMSE shrinks at the $\sqrt{n}$ rate predicted by Theorem 4 of Lewis and Syrgkanis (2021). Coverage of the influence-function confidence intervals is uniformly close to the nominal 95% level across the sample-size grid.

Table 2: Sim 1, dynamic DML on a two-stage SNMM. Bias, root-mean-square error, and 95%-CI coverage for each estimator at $n = 4000$, over 200 Monte Carlo replications. True parameters $\psi_1^* = 1.0$, $\psi_2^* = 0.5$.

	$\psi_1^* = 1.00$			$\psi_2^* = 0.50$		
Method	Bias	RMSE	Cov	Bias	RMSE	Cov
Naive OLS	-0.191	0.197	0.01	+0.933	0.934	0.00
MSM/IPTW	+0.019	0.103	0.88	+0.146	0.192	0.73
Dynamic DML	-0.005	0.046	0.96	-0.001	0.018	0.93

Sim 2: causal bandits on the parallel-bandit family

The data-generating process matches the parallel-bandit setting of Lattimore et al. (2016, §3). The reward Y depends on $N = 50$ binary parents $X_1, \ldots, X_N$ through a single high-leverage coordinate: $\mathbb{E}[Y \mid X] = 0.5 + \epsilon$ if $X_w = 1$ and $0.5 - \epsilon$ otherwise, with $\epsilon = 0.3$ and the index w drawn uniformly at random per replication. The action set has $2N + 1 = 101$ arms: the empty intervention do() and each binary intervention $\mathrm{do}(X_i = j)$. Under the

[7] Sim 1 source: `ch11_rl_for_ci/sims/dynamic_dml_snmm.py`.

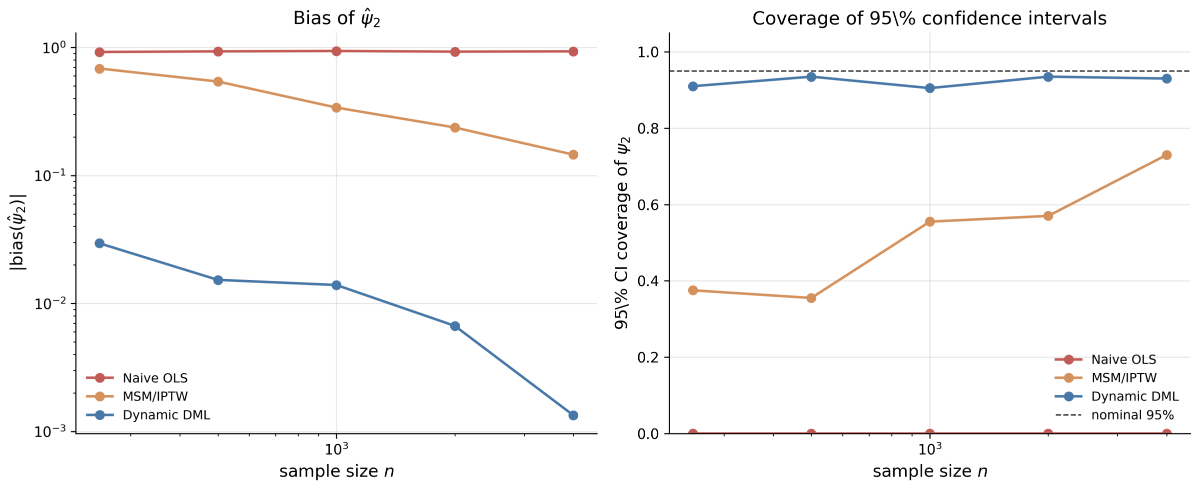

Figure 2: Sim 1, dynamic DML on a two-stage SNMM. Left panel: absolute bias of $\hat{\psi}_2$ versus sample size n (log-log). Right panel: empirical coverage of nominal 95% confidence intervals on ψ_2 versus n. Dotted reference line is the nominal coverage level. Each point averages 200 Monte Carlo replications under the data-generating process in Equation (18).

observational distribution each parent X_i is Bernoulli with mean q_i; the graph-derived hardness

$$m(q) = \min_{\tau \in \{2,\dots,N\}} \max\Big(\tau,\ \Big|\{i : \min(q_i, 1-q_i) < 1/\tau\}\Big|\Big) \in [2, N] \tag{19}$$

is controlled by setting exactly m^* coordinates of q to the extreme value 0.05 and the remainder to 0.5, giving $m(q) = m^*$ by construction.

Three algorithms are compared. *Successive Reject* (Audibert and Bubeck 2010, as in Lattimore et al., 2016) is the graph-blind best-arm-identification baseline that allocates the budget across all $2N + 1$ arms using the standard logarithmic schedule. *Lattimore Algorithm 1* spends $T/2$ rounds on do() to estimate the parent propensities and the conditional means $\hat{\mathbb{P}}(Y \mid X_i = j)$ that identify each direct-causal arm in the parallel graph, then allocates the remaining $T/2$ rounds uniformly across the unbalanced arms whose direct-observation probability falls below $1/\hat{\tau}^*$. *Causal Thompson sampling* (TS_C) is the algorithm of Bareinboim et al. (2015) on the greedy-casino MABUC instance described below: it seeds Beta posteriors from observational data via the consistency axiom, then at each step samples from the agent's intuition-conditional posterior and applies the Regret Decision Criterion. The vanilla Thompson sampling baseline ignores intuition entirely and tracks a single Beta posterior per arm.

For the simple-regret experiments two thousand Monte Carlo replications are run on a grid of hardness values $m^* \in \{2, 8, 24, 48\}$ at horizon $T = 400$, and on a grid of horizons $T \in \{100, 200, 400, 800, 1600\}$ at fixed hardness $m^* = 2$. For the MABUC experiment five hundred replications are run at $T = 1000$ using the greedy-casino payoffs of Bareinboim et al. (2015, Table 1): under each of two unobserved-confounder configurations $X \in \{0, 1\}$, following the agent's natural intuition yields payoff 0.1 and going against it yields 0.5. The observational distribution and the experimental distribution both average out to $0.5 \times 0.1 + 0.5 \times 0.5 = 0.3$ per arm, making standard Thompson sampling unable to distinguish the arms in expectation.

Table 3 and Figure 3 report the results.[8] The graph-blind Successive Reject algorithm's simple regret is essentially flat in m^* at approximately 0.28, very close to the theoretical $\sqrt{N/T} \cdot \epsilon \approx 0.106$ scaled by the gap; it is bottlenecked by the cost of exploring all $2N + 1$ opaque arms. The graph-aware algorithm achieves regret 0.024 at $m^* = 2$ and rising approximately as $\sqrt{m^*/T}$ across the grid, in line with Theorem 1 of Lattimore et al. (2016). In the horizon-sweep panel at fixed $m^* = 2$, Lattimore Algorithm 1's regret falls from 0.13 at $T = 100$ to less than 0.001 at $T = 1600$, while Successive Reject's regret falls from 0.31 to only 0.14 over the same range. In the greedy-casino MABUC panel, vanilla Thompson sampling accumulates cumulative regret 200.5 at $T = 1000$ (linear in T, as predicted), while causal Thompson sampling accumulates only 0.66 over the same horizon, a ratio of approximately 305×.

Table 3: Sim 2, causal bandits on the parallel-bandit family. Simple regret (mean, standard error in parentheses) for the graph-blind Successive Reject baseline and the graph-aware Lattimore Algorithm 1 across the graph-hardness grid, at horizon $T = 400$, $N = 50$ parents, gap $\epsilon = 0.3$, two thousand Monte Carlo replications per cell.

Method	$m = 2$	$m = 8$	$m = 24$	$m = 48$
Successive Reject	0.281 (0.002)	0.281 (0.003)	0.295 (0.005)	0.318 (0.006)
Lattimore Alg 1	0.024 (0.002)	0.073 (0.003)	0.125 (0.004)	0.071 (0.004)

Both simulations corroborate the theoretical claims of the underlying papers. Sim 1 shows that the Lewis-Syrgkanis recursive residualisation moment delivers the $\sqrt{n}$ asymptotic normality predicted by Theorem 4 of Lewis and Syrgkanis (2021), even though the state is high-dimensional, the propensities are confounded, and naive panel regression

[8] Sim 2 source: `ch10b_rl_for_ci/sims/causal_bandit_parallel.py`.

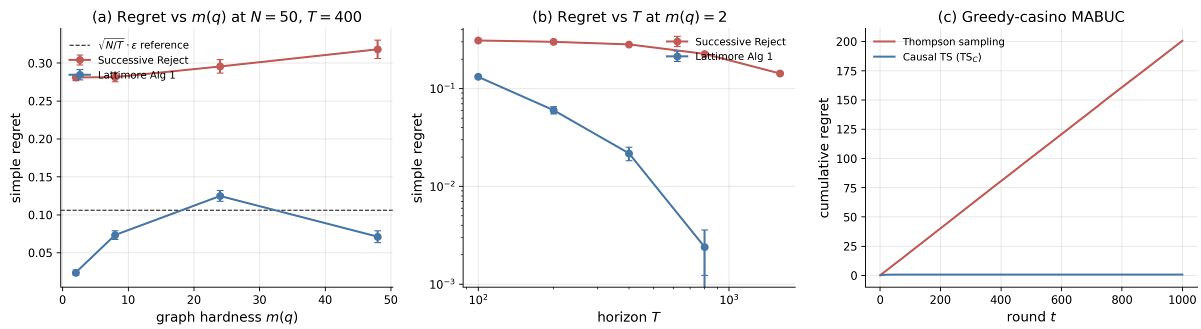

Figure 3: Sim 2, causal bandits. (a) Simple regret versus graph hardness $m(q)$ at fixed horizon $T = 400$ and parent count $N = 50$; the dotted reference line is the theoretical $\sqrt{N/T} \cdot \epsilon$ floor for graph-blind algorithms. (b) Simple regret versus horizon T at fixed hardness $m(q) = 2$, log-log axes. (c) MABUC greedy-casino panel under the unobserved-confounder instance of Bareinboim et al. (2015, Table 1): vanilla Thompson sampling grows linearly because the observational and interventional marginals are both flat, while causal Thompson sampling remains bounded by conditioning on the agent's natural intuition. Error bars and shaded bands are 95% confidence intervals; two thousand Monte Carlo replications for (a)–(b), five hundred for (c).

with the same nuisance learners is biased. Sim 2 shows the matching upper and lower bounds of Lattimore et al. (2016, Theorems 1–2) on simple regret hold tightly across the hardness grid, and that the MABUC counterfactual reasoning of Bareinboim et al. (2015) converts a hopeless bandit problem into an almost-trivial one. The chapter's broader thesis is that these are the same machine learning machinery applied at the design stage and the inference stage of dynamic causal-inference problems, brought to bear under the identification assumptions catalogued in Section 0.1.

0.7 Discussion

The five sections of this chapter share one underlying claim. Dynamic causal inference and reinforcement learning are not separate enterprises that happen to use similar notation; they are the same enterprise applied to different data sources. The recursion in Section 0.1 is Bellman's. The Lewis-Syrgkanis sequential residualisation is Robins's g-estimation with cross-fitting added. The Sakaguchi backward-induction AIPW is the offline-RL Q-learning update with doubly-robust scoring. The Lattimore causal bandit is the multi-armed bandit with the action set replaced by interventions on a known graph. The Kasy-Sautmann exploration sampling is Thompson sampling reshaped for a different objective function. In each case the move is not to invent a new algorithm but to recognize that an algorithm developed in one tradition addresses a question that arose in the other tradition.

Three implications follow for the empirical worker.

First, the rate of convergence of an estimator is determined by the orthogonality of the moment that defines it, not by the field in which the moment was first written down. The orthogonal statistical learning theory of Foster and Syrgkanis (2023) unifies the dynamic-DML rate calculations of Section 0.2 with the policy-learning regret bounds of Section 0.3 and the post-bandit-inference asymptotic normality of Section 0.5. The common requirement is that nuisance estimation errors enter the target estimator at second order; the common consequence is that any sufficiently fast machine learner can be used for the nuisance components without sacrificing the parametric rate on the structural target. This is the modern semiparametric efficiency story, transported into reinforcement learning territory.

Second, identification is prior to estimation. The recursions in this chapter all rely on sequential ignorability, positivity, and consistency. When those assumptions hold by design (a sequential multiple-assignment randomized trial, an adaptive field experiment with a known assignment mechanism, a known causal graph), the chapter's techniques deliver $\sqrt{n}$ inference on dynamic causal quantities. When they fail, the recursions still run but they no longer estimate causal quantities; the proxy, instrument, sensitivity, and mediator strategies of Chapter **??** are the recovery routes. The economist's instinct to ask first what identifies the parameter of interest, then what estimates it, applies unchanged here.

Third, the bridge runs in both directions. Several of the most useful results in modern reinforcement learning are descendants of biostatistical and econometric reasoning rather than purely computer-scientific. The advantage doubly-robust estimator of Nie et al. (2021) is the dynamic generalization of the Athey-Wager policy-learning estimator. The adaptively-weighted AIPW of Hadad et al. (2021) is Robins-style augmented inverse-propensity weighting reweighted for adaptive data collection. The recursive Riesz representers of Chernozhukov et al. (2023) are an automation of the locally-robust correction that Robins introduced in g-estimation. The traffic of ideas is two-way, and a chapter framed as "reinforcement learning for causal inference" could equally be read as "causal inference recovers, validates, and extends reinforcement learning under identification."

Open problems remain in several directions. The dynamic-DML rate of Lewis and

Syrgkanis (2021) is established for parametric structural nested mean models with linear blips; extensions to fully nonparametric blip functions, to the high-stake combination of large state spaces and adaptively collected data, and to settings where the policy class itself is learned from the data (rather than pre-specified) are active research topics. The matching minimax bound of Lattimore et al. (2016) applies to the parallel-bandit setting; for general directed acyclic graphs the upper bound is established but the lower bound is not. The adaptively-weighted AIPW of Hadad et al. (2021) assumes a known assignment mechanism; relaxing this to allow the analyst to recover valid inference from contextual bandit data collected by a black-box system is the contribution of Bibaut et al. (2021), but the corresponding practical guidance remains to be developed. Each of these open problems is, at the methodological level, a question about how to import additional machine-learning machinery into a causal-inference recursion without losing the orthogonality property that makes the rate work.

A final remark on practice. The PAD India field experiment of Kasy and Sautmann (2021) and the Tennessee Project STAR re-analysis of Sakaguchi (2024) are the chapter's clearest demonstrations that the theory has practical purchase: a reinforcement-learning algorithm at the design stage with a causal-inference estimator at the inference stage produces valid welfare gains on real participants in real settings. The chapter's broader message is that this combination is no longer exotic. It is the natural next step once one accepts that dynamic causal inference and reinforcement learning are doing the same thing under different names, and that the modern machine-learning toolkit, applied carefully, can serve both at once.

References

Susan Athey and Stefan Wager. Policy learning with observational data. *Econometrica*, 89(1):133–161, 2021.

Leemon C. Baird. Advantage updating. Technical Report WL-TR-93-1146, Wright Laboratory, Wright-Patterson AFB, 1993.

Elias Bareinboim, Andrew Forney, and Judea Pearl. Bandits with unobserved confounders: A causal approach. In *Advances in Neural Information Processing Systems*, volume 28, pages 1342–1350, 2015.

Richard Bellman. *Dynamic Programming*. Princeton University Press, 1957.

Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*. Athena Scientific, 1996.

Aurélien Bibaut, Antoine Chambaz, Maria Dimakopoulou, Nathan Kallus, and Mark van der Laan. Post-contextual-bandit inference. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

Victor Chernozhukov, Whitney Newey, Rahul Singh, and Vasilis Syrgkanis. Automatic debiased machine learning for dynamic treatment effects and general nested functionals. *arXiv preprint arXiv:2203.13887*, 2023.

Damien Ernst, Pierre Geurts, and Louis Wehenkel. Tree-based batch mode reinforcement learning. *Journal of Machine Learning Research*, 6:503–556, 2005.

Nan Fang, Wei Gong, and Guiliang Liu. Offline inverse constrained reinforcement learning for safe-critical decision making in healthcare. *arXiv preprint arXiv:2410.07525*, 2024.

Dylan J. Foster and Vasilis Syrgkanis. Orthogonal statistical learning. *Annals of Statistics*, 51(3):879–908, 2023.

Vitor Hadad, David A. Hirshberg, Ruohan Zhan, Stefan Wager, and Susan Athey. Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the National Academy of Sciences*, 118(15):e2014602118, 2021.

Takanori Ida, Takunori Ishihara, Koichiro Ito, Daido Kido, Toru Kitagawa, Shosei Sakaguchi, and Shusaku Sasaki. Choosing who chooses: Selection-driven targeting in energy rebate programs. Technical Report 32561, National Bureau of Economic Research, 2024.

Maximilian Kasy and Anja Sautmann. Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132, 2021.

Pramod Kaushik, Sneha Kummetha, Perusha Moodley, and Raju S. Bapi. A conservative Q-learning approach for handling distribution shift in sepsis treatment strategies. *arXiv preprint arXiv:2203.13884*, 2022.

Taylor W. Killian, Marzyeh Ghassemi, and Shalmali Joshi. Counterfactually guided off-policy transfer in clinical settings. In *Conference on Health, Inference, and Learning (CHIL)*, 2022.

Toru Kitagawa and Aleksey Tetenov. Who should be treated? Empirical welfare maximization methods for treatment choice. *Econometrica*, 86(2):591–616, 2018.

Flemming Kondrup, Thomas Jiralerspong, Elaine Lau, Nathan de Lara, Jacob Shkrob, My Duc Tran, Doina Precup, and Sumana Basu. Towards safe mechanical ventilation treatment using deep offline reinforcement learning. In *AAAI Conference on Artificial Intelligence (Innovative Applications)*, 2023.

Finnian Lattimore, Tor Lattimore, and Mark D. Reid. Causal bandits: Learning good interventions via causal inference. In *Advances in Neural Information Processing Systems*, volume 29, pages 1181–1189, 2016.

Greg Lewis and Vasilis Syrgkanis. Double/debiased machine learning for dynamic treatment effects. In *Advances in Neural Information Processing Systems*, volume 34, pages 22695–22707, 2021.

Daniel J. Luckett, Eric B. Laber, Anna R. Kahkoska, David M. Maahs, Elizabeth Mayer-Davis, and Michael R. Kosorok. Estimating dynamic treatment regimes in mobile health using V-learning. *Journal of the American Statistical Association*, 115(530): 692–706, 2020.

Zhiyao Luo, Mingcheng Zhu, Fenglin Liu, Jiali Li, Yangchen Pan, Jiandong Zhou, and Tingting Zhu. DTR-Bench: An in silico environment and benchmark platform for reinforcement learning based dynamic treatment regime. *arXiv preprint arXiv:2405.18610*, 2024. Under review.

S. A. Murphy. Optimal dynamic treatment regimes. *Journal of the Royal Statistical Society, Series B*, 65(2):331–355, 2003.

Xinkun Nie, Emma Brunskill, and Stefan Wager. Learning when-to-treat policies. *Journal of the American Statistical Association*, 116(533):392–409, 2021.

Doina Precup, Richard S. Sutton, and Satinder P. Singh. Eligibility traces for off-policy policy evaluation. In *Proceedings of the 17th International Conference on Machine Learning (ICML)*, pages 759–766, 2000.

James M. Robins. A new approach to causal inference in mortality studies with a sustained exposure period — application to control of the healthy worker survivor effect. *Mathematical Modelling*, 7(9-12):1393–1512, 1986.

James M. Robins. Optimal structural nested models for optimal sequential decisions. In D. Y. Lin and P. J. Heagerty, editors, *Proceedings of the Second Seattle Symposium in Biostatistics*, volume 179 of *Lecture Notes in Statistics*, pages 189–326. Springer, New York, 2004.

James M. Robins, Miguel Á. Hernán, and Babette Brumback. Marginal structural models and causal inference in epidemiology. *Epidemiology*, 11(5):550–560, 2000.

Luca F. Roggeveen, Ali el Hassouni, Harm-Jan de Grooth, Armand R. J. Girbes, Mark Hoogendoorn, and Paul W. G. Elbers. Reinforcement learning for intensive care medicine: Actionable clinical insights from novel approaches to reward shaping and off-policy model evaluation. *Intensive Care Medicine Experimental*, 12(1):32, 2024. doi: 10.1186/s40635-024-00614-x.

Donald B. Rubin. Bayesian inference for causal effects: The role of randomization. *The Annals of Statistics*, 6(1):34–58, 1978.

Shosei Sakaguchi. Policy learning for optimal dynamic treatment regimes with observational data. *arXiv preprint arXiv:2404.00221*, 2024. Revise-and-resubmit at *Journal of Econometrics*.

Phillip J. Schulte, Anastasios A. Tsiatis, Eric B. Laber, and Marie Davidian. Q- and A-learning methods for estimating optimal dynamic treatment regimes. *Statistical Science*, 29(4):640–661, 2014.

Chengchun Shi, Shikai Luo, Yuan Le, Hongtu Zhu, and Rui Song. Statistically efficient advantage learning for offline reinforcement learning in infinite horizons. *Journal of the Royal Statistical Society, Series B*, 2024. Accepted; arXiv:2202.13163.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. The MIT Press, 2nd edition, 2018.

Christopher J. C. H. Watkins and Peter Dayan. Q-learning. *Machine Learning*, 8(3-4): 279–292, 1992. doi: 10.1007/bf00992698.

Changchang Yin, Ruoqi Liu, Jeffrey Caterino, and Ping Zhang. Deconfounding actor-critic network with policy adaptation for dynamic treatment regimes. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022.

Zhengyuan Zhou, Susan Athey, and Stefan Wager. Offline multi-action policy learning: Generalization and optimization. *Operations Research*, 71(1):148–183, 2023.